GUAN HUANG

APPLIED ML ENGINEER

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Applied ML Engineer with **4+ years** taking ML systems from design to deployment, focused on rigorous evaluation, retrieval and inference efficiency. Led teams of 4–5 to ship production LLM agents and **Railtracks**, an open-source agent framework. MSc in Applied Computing (AI), University of Toronto.

4+	10+	200+	3
YEARS OF APPLIED ML	PRODUCTION AGENTS SHIPPED	GITHUB STARS	PATENTS PENDING

EXPERIENCE

May 2023 – Sep 2025

Applied ML Engineer / Railtown AI Technology Inc.

- Led a team of five to ship Conductr, a **DevOps agent** integrating Azure, Jira and GitHub that triages work items, monitors stack traces and answers questions over source code and Git history. Won **90% of blind pairwise comparisons** against a leading commercial coding assistant (two-sided binomial, $p<0.01$).
- Designed and built **Railtracks**, the company's **open-source agent framework**.
- Built AEval, an automated **AI expert system evaluation** tool (accuracy, hallucination, security, fairness), now embedded in **hundreds of deployed products**.
- Worked with Mila researchers on stack-trace NLP models and ran product user studies.

Oct 2024 – Apr 2025

Undergraduate Researcher / UBC SALT Lab – ECE Department

- **Control-flow guided test generation:** used execution paths to guide language-agnostic test generation, raising branch coverage from 88% to 93% with Llama 3.
- Proposed and led **SoC visual schematic generation with VLMs**: working with HDL hardware files through a vision-language model with an entity database and feedback loop.

May 2022 – May 2023

Data Scientist / Railtown AI Technology Inc.

- Developed a **multi-model ensemble framework** combining language model output embeddings: 91% user preference in A/B testing (two-sided binomial, $p<0.001$).
- Rebuilt **root cause analysis** with an LLM (F1 from 0.71 to 0.85), plus its fine-tuning pipeline.

SELECTED PROJECTS

Mar 2024 – Present

Creator & Maintainer / Railtracks OPEN SOURCE

- Plain-Python agent framework where each part (loop, tools, limits, run record) is an object you can read and replace. **200+ GitHub stars**, about **1,500 downloads a month**.
- Any function becomes a tool with one decorator, agents call other agents as tools, and every run is recorded and replayed locally in a visualizer.
- Co-authored its public report and ran the Railtracks x Amii hackathon.

Jan 2025 – Present

ML Team Lead / FeatherRing COMMUNITY PROJECT

- Lead the Intelligence Core team of this invite-only project: a **multi-agent system that operates the computer** and understands all media types.
- Trained an **entity-based context compression** model (agentic entity extraction, selective summarization, memory retrieval) and a **steerable LLM** that takes instructions while it generates.

Sep 2026 – Present

Author & Maintainer / EvalRing OPEN SOURCE

- **Evaluation tool** that ranks models, prompts or pipelines on your task, reproducibly.

Oct 2026 – Present

Designer & Developer / NewsRing OPEN SOURCE

- **Daily AI news agent** on Railtracks that picks only from sources it collected.

Apr 2025 – Sep 2025

Advisor/Engineer / ClickCut PARTNERING WITH NUCHAIN

- Led a group of six building an **AI video editor** that plans and produces hour-long videos, with its deployment, fine-tuning and data pipelines.

EDUCATION

Sep 2025 – Nov 2026

MSc in Applied Computing – Artificial Intelligence / University of Toronto

Sep 2019 – Apr 2025

BASc in Computer Engineering / University of British Columbia

Top 1% (grade 93, honours); Trek Excellence, PMC-Sierra and James Yan scholarships

LEADERSHIP

Sep 2025 – Present

Vice President / UofT CS Graduate Student Society

- Represent 600+ CS graduate students in governance and outreach to 20+ firms.

May 2022 – Aug 2025

Head Secretary / IEEE UBC Student Chapter

- Grew member enrollment by 150% and public events by 75%; led recruitment.

PATENTS & PUBLICATIONS

PATENTS	Three pending: ticket generation, RCA, release notes and DevOps knowledge copilots.
PUBLICATIONS	*Railtracks Agentic Framework* report (co-author); *MapRing*, *LORAG*, *VSFL* (first author), 2025.

SKILLS

LANGUAGES	Python (**PyTorch, JAX**), JS/TS, C++ (CUDA), Java, Dart
ML	**LLM**, VLM, **evals**, RAG, multi-agent systems, fine-tuning, AI security, human-AI interaction
MLOPS & TOOLS	CI/CD, Docker/K8s, AzureML, vLLM, LangChain, Google ADK, Railtracks, vector stores
SYSTEMS	Full-stack, API design, distributed computing, GPU programming, Verilog/FPGA, ROS

FULL CV — Awards, competitions, research, more projects and events: coronring.github.io/resume